# XIAMEN UNIVERSITY MALAYSIA

## MID-TERM EXAMINATION

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| Course Code: | COM108 | | | |
| Course Name: | Engineering Maths (I) | | | |
| Question Paper Setter: | Liu Meifeng | | | |
| Academic Session: | 2021/09 | Question Paper: | A ☑ | B ☐ |
| Total No. of Pages: | 4 | Time Allocated: | 2 hours | |
| Additional Materials: | - | | | |
| Apparatus Allowed: | Calculator | | | |

**INSTRUCTIONS TO CANDIDATES**

1. There are FOUR questions in this paper, please answer ALL questions in this paper.
2. Read the above information carefully to ensure you have the correct and complete question paper.
3. Please follow the requirement of each section and write down all the corresponding answers on the answer book provided.
4. Please note that presentation of solutions (clarity, coherence, conciseness, and completeness) is important. Show your work and organize your solution. Answer without proper justification may receive less, or even zero credit.
5. Communication between candidates in any means is forbidden. Answers must be entirely individual candidate's independent effort. If you are found sharing your solutions with other candidates, or suspected of doing so, you would be penalized accordingly.

**DO NOT TURN OVER THIS PAGE UNTIL INSTRUCTED TO DO SO.**

(Student ID: _____ Full Name: _____ )

**Question 1 (25 marks)**

(a) A function is defined below. [5 marks]

$$f(x) = \begin{cases} 3 - x, & \text{if } x \leq -2 \\ x^3 & \text{if } x > -2 \end{cases}.$$

Evaluate $f(-3)$, $f(-2)$, and $f(2)$ and sketch the graph.

(b) Find the function $f(x)$ that satisfies the following conditions. [5 marks]

(i) $f(x)$ is a polynomial of degree 3

(ii) the value of $f(x)$ when $x = 3$ is 72

(iii) the three roots of $f(x) = 0$ is -3, 0 and 4

(c) Find $f \circ g \circ h$. [5 marks]

$$f(x) = \tan(x), \qquad g(x) = \frac{x}{x-2}, \qquad h(x) = \sqrt[3]{x}.$$

(d) Given the function $h(x) = \sin|2x|$.

(i) Sketch the graph of the function. [2 marks]

(ii) Determine whether the function is even, odd, or neither. Explain why. [3 marks]

(e) Let $f(x) = \ln(e^x - 2)$.

(i) Find the domain and range of $f(x)$. [2 marks]

(ii) Find the inverse function $f^{-1}(x)$, state its domain and range. [3 marks]

**Question 2 (25 marks)**

(a) Find the limit if it exists. [5 marks]

$$\lim_{x\to 0.1^-} \frac{10x-1}{|10x^3-x^2|}$$

(b) Let $f(x)$ be a continuous function defined by

$$f(x)=\begin{cases} 3+x^2 & \text{if } x\le 0 \\ a-x & \text{if } 0<x\le 4. \\ -(x-b)^2 & \text{if } x>4 \end{cases}$$

(i) Determine the value of $a$. [2 marks]

(ii) Determine the value of $b$. [3 marks]

(c) Evaluate each of the following limits WITHOUT using L'Hospital's Rule.

(i) $\lim_{t\to 0} \frac{1-\cos\ t}{\sin t}$ [2 marks]

(ii) $\lim_{x\to -3} \frac{\left(\sqrt{1-x}-2\right)}{x+3}$ [2 marks]

(iii) $\lim_{n\to\infty} \left(1-\frac{3}{n}\right)^n$ [2 marks]

(iv) $\lim_{x\to -\infty} \frac{10x^2-3x+1}{5x^2-4x-1}$ [2 marks]

(v) $\lim_{x\to -\infty} 2x+\sqrt{4x^2-2x+3}$ [2 marks]

(d) Evaluate the following limit. [5 marks]

$$\lim_{x\to 0^+} (\cos x)^{x^{-1}}$$

**Question 3 (25 marks)**

(a) If $F(x) = f(g(x))$, where $f(-2) = 6$, $f'(-2) = 3$, $f'(-1) = 4$, $g(-1) = -2$, and $g'(-1) = 6$, find $F'(-1)$. [5 marks]

(b) Let the function be defined as

$$y = f(x) = x^8 \ln x$$

(i) Find $y'$ and $y''$. [2 marks]

(ii) Find $y^{(9)}$. [3 marks]

(c) Find $dy/dx$ by implicit differentiation. [5 marks]

$$1 + 7x = \sin(xy^2)$$

(d) Find an equation of the tangent line to the following curve at the given point. [5 marks]

$$y = 8e^x \cos(x), \qquad (0,8)$$

(e) The edge of a cube was found to be 15 cm with a possible error in measurement of 0.2 cm. Use differentials to estimate the maximum possible error in computing the volume of the cube and the surface area of the cube. (Express your answers fraction form.) [5 marks]

**Question 4 (25 marks)**

(a) A particle moves according to a law of motion $s = f(t)$, [10 marks]

$$f(t) = t^3 - 9t^2 + 24t, \qquad t > 0$$

where $t$ is measured in seconds and $s$ in feet.

(i) Find the velocity and acceleration as functions of $t$.

(ii) Find the acceleration of the moving particle after 8 second.

(iii) Find the total distance travelled of the moving particle during the first 8 seconds.

(b) Consider the equation below. [5 marks]

$$f(x) = \frac{\ln x}{x}$$

(i) Find the horizontal and vertical asymptotes, if any.

(ii) Find the intervals of concavity.

(iii) Find the inflection point, if any.

(c) Find $f(x)$, if [5 marks]

$$f''(x) = 8 + 6\cos x + 36x^2, \qquad f(0) = 3, \qquad f(1) = 15.$$

(d) Find the dimensions of a rectangle with perimeter 80 m whose area is as large as possible. [5 marks]

**- END OF PAPER -**